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EDUCATION

B.S. in Computer Systems Engineering, Arizona State University **December 2025**

- GPA: 3.11

Essential Courses: Computer Architecture, Design and Synthesis of Digital Hardware, Embedded Microprocessor Systems, Introduction to Software Engineering, Computer Networks, Operating Systems

PROFESSIONAL EXPERIENCE

Software Intern – DSRL, India **July 2024 - September 2024**

- Developed Dockerized ROS 2 IMU nodes on Raspberry Pi for real-time sensor data collection
- Integrated I2C devices across Docker networks for robust data transmission

Research Intern — University of South Florida (Remote) **May 2023 - Jan 2024**

- Built Python-based data pipelines to clean, validate, and visualize gas analysis data
- Cleaned, visualized, and documented lab data for technical reporting and presentations

Programming Assistant Intern, PIS IT Solutions Pvt. Ltd, India **June 2022-Aug 2022**

- Built a mock test web platform using MySQL, HTML/CSS, and JavaScript
- Wrote and optimized SQL queries for backend data retrieval and user authentication
- Improved debugging workflow, reducing resolution time by 15%

PROJECTS

Socket Project, CSE 434(Computer Networks): **Fall 2024**

- Developed a multi-threaded peer-to-peer application in Python, implementing socket programming to manage server and client communication
- Achieved robust tracking of player and game states while maintaining a detailed version control history in GitHub, improving collaboration and traceability.

Speed Control System, CSE 325 (Embedded Systems): **Fall 2024**

- Designed and implemented a control algorithm in C to maintain motor speed under variable load conditions using embedded systems.
- Developed and tested hardware setups involving GPIO registers and motor encoders, ensuring consistent performance across diverse scenarios.
- Prepared a detailed report and conducted a demo to showcase project implementation and results.

Project 6, CSE330(Operating Systems): **Fall 2024**

- Developed a Linux kernel module for disk caching using dm-cache, implementing and testing in a virtualized environment on UTM to simulate enterprise-level storage systems.

Role-Based Help System(introduction to Software Engineering) **Fall 2024**

- Designed and implemented a secure, role-based help system using JavaFX, enabling personalized content delivery and robust access controls for students, admins, and instructional teams.
- Developed and optimized search functionality with keyword tagging and user-specific filters, enhancing usability and content relevance.
- Created administrative tools for managing articles, including backup, restore, and role-based access features, ensuring seamless data management across semesters.

Digital Circuit Design and FPGA Prototyping **Fall 2023**

- Designed and simulated digital circuits using Verilog, implementing and verifying functionality through waveform analysis.
- Utilized FPGA platforms to prototype digital designs and validate performance against timing requirements.

Automated Edge Deployment Platform – Frontend & IaaS Integration
ARED Group Inc · Capstone Project **Spring 2025**

- Designed and implemented automated Tailscale network registration for edge devices using Python and API key authentication, streamlining zero-touch deployment.
- Dockerized registration system for scalable deployment across distributed edge nodes.
- Integrated Prometheus monitoring stack, including Redis Exporter and Node Exporter, to enable system, service, and application-level visibility.
- Configured /metrics endpoints and created custom alert rules for detecting downtime, resource overuse, and anomalies.
- Enhanced platform reliability and observability for distributed application management in resource-constrained environments.
- Added three Prometheus-like features to optimize the existing Python monitoring system.
- Created a new pop-up notification panel for the front-end dashboard